

Analysis and Interpretation of Financial Statements — Complete Notes

Grade 12

Accounting

CAPS

What you'll learn

- Why we calculate and compare financial ratios
- The profitability ratios: gross profit margin, net profit margin and return on equity
- The liquidity ratios: current ratio and acid-test ratio
- The solvency ratio: debt-equity ratio
- The efficiency ratios: stock turnover, debtors collection period and creditors payment period
- How to write a proper interpretation comment, not just a number
- The errors that lose marks in the Grade 12 examination

1. Why ratios?

A single figure tells you very little. A net profit of R300 000,00 is impressive for a small business but weak for a giant one. Ratios put the figures into context by comparing them with sales, with equity, with a previous year or with another business. In the examination you normally compare the current year with the previous year and comment on whether the ratio has improved or weakened, why it changed, and what the consequence is.

The standard method is a four-step answer:

1. **Give the ratio** as a figure or percentage with the correct unit.
2. **Compare** it with the previous year (better, worse, or stable).
3. **Explain the cause** using the underlying figures.
4. **State the consequence** for the business.

2. Ratio formulas

Ratio	Formula	What it measures
Gross profit margin	Gross profit ÷ Sales × 100	Profitability of trading before expenses
Net profit margin	Profit after tax ÷ Sales × 100	Overall profitability of every rand of sales
Return on equity	Profit after tax ÷ Average equity × 100	Return earned for the shareholders
Current ratio	Current assets ÷ Current liabilities	Ability to pay short-term debts
Acid-test ratio	(Current assets – Inventories) ÷ Current liabilities	Ability to pay short-term debts without selling stock
Debt-equity ratio	Non-current liabilities ÷ Equity	Financial risk from borrowing
Stock turnover	Cost of sales ÷ Average stock	How many times stock is sold and replaced
Debtors collection period	Trade receivables ÷ Credit sales × 365	Average days to collect money from debtors
Creditors payment period	Trade payables ÷ Credit purchases × 365	Average days taken to pay creditors

Average stock = (opening stock + closing stock) ÷ 2. Average equity = (opening equity + closing equity) ÷ 2.

3. Worked example 1 — profitability ratios

Case: Lulama Ltd for the year ended 30 June 2026.

Extract	2026	2025
Sales	2 000 000,00	1 800 000,00
Cost of sales	1 200 000,00	1 116 000,00
Gross profit	800 000,00	684 000,00
Profit after tax	300 000,00	240 000,00
Equity	1 200 000,00	1 000 000,00

Gross profit margin: $800\,000,00 \div 2\,000\,000,00 \times 100 = 40,0\%$. In 2025 it was $684\,000,00 \div 1\,800\,000,00 \times 100 = 38,0\%$. It improved, which means the company earned more gross profit on every rand of sales — possibly through a higher mark-up or better buying.

Net profit margin: $300\,000,00 \div 2\,000\,000,00 \times 100 = 15,0\%$. In 2025 it was $240\,000,00 \div 1\,800\,000,00 \times 100 = 13,3\%$. It improved, so the company kept more of each rand of sales as profit after all expenses and tax.

Return on equity: average equity = $(1\,200\,000,00 + 1\,000\,000,00) \div 2 = 1\,100\,000,00$. $300\,000,00 \div 1\,100\,000,00 \times 100 = 27,3\%$. Shareholders earned a return of 27,3 cents for every rand of equity, which is a strong return.

Ratio	2026	2025	Trend
Gross profit margin	40,0%	38,0%	Improved
Net profit margin	15,0%	13,3%	Improved
Return on equity	27,3%	—	Computed

4. Worked example 2 — liquidity and solvency ratios

Using the balance sheet extracts for Lulama Ltd:

Extract	2026	2025
Inventories	200 000,00	160 000,00
Trade receivables	350 000,00	300 000,00
Cash and cash equivalents	50 000,00	40 000,00
Current assets	600 000,00	500 000,00
Current liabilities	250 000,00	200 000,00
Non-current liabilities	300 000,00	350 000,00
Equity	1 200 000,00	1 000 000,00

Current ratio: $600\,000,00 \div 250\,000,00 = 2,4 : 1$. In 2025 it was $500\,000,00 \div 200\,000,00 = 2,5 : 1$. The ratio weakened slightly but is still above the ideal of $2 : 1$, so the company can comfortably pay its short-term debts.

Acid-test ratio: $(600\,000,00 - 200\,000,00) \div 250\,000,00 = 400\,000,00 \div 250\,000,00 = 1,6 : 1$. In 2025 it was $(500\,000,00 - 160\,000,00) \div 200\,000,00 = 340\,000,00 \div 200\,000,00 = 1,7 : 1$. It remains above the ideal of $1 : 1$, which is healthy.

Debt-equity ratio: $300\,000,00 \div 1\,200\,000,00 = 0,25 : 1$. In 2025 it was $350\,000,00 \div 1\,000\,000,00 = 0,35 : 1$. The company relies less on borrowed money than before, which lowers financial risk.

Ratio	2026	2025	Trend
Current ratio	2,4 : 1	2,5 : 1	Slightly weaker, still sound
Acid-test ratio	1,6 : 1	1,7 : 1	Slightly weaker, still sound
Debt-equity ratio	0,25 : 1	0,35 : 1	Lower risk

5. Worked example 3 — efficiency ratios

Additional information for Lulama Ltd: all sales are on credit, credit purchases for 2026 were R1 250 000,00, credit purchases for 2025 were R1 100 000,00, trade payables are R200 000,00 (2026) and R170 000,00 (2025), and stock on 1 July 2024 was R140 000,00.

Stock turnover (2026): average stock = $(200\,000,00 + 160\,000,00) \div 2 = 180\,000,00$. $1\,200\,000,00 \div 180\,000,00 = 6,7$ times. In 2025 average stock = $(160\,000,00 + 140\,000,00) \div 2 = 150\,000,00$, so $1\,116\,000,00 \div 150\,000,00 = 7,4$ times. Stock now moves more slowly, which could mean weaker demand or too much stock on hand.

Debtors collection period (2026): $350\,000,00 \div 2\,000\,000,00 \times 365 = 63,9 \text{ days}$. In 2025 it was $300\,000,00 \div 1\,800\,000,00 \times 365 = 60,8 \text{ days}$. Debtors are taking longer to pay, which ties up cash. The company may need to tighten credit control.

Creditors payment period (2026): $200\,000,00 \div 1\,250\,000,00 \times 365 = 58,4 \text{ days}$. In 2025 it was $170\,000,00 \div 1\,100\,000,00 \times 365 = 56,4 \text{ days}$. The company is taking slightly longer to pay its suppliers. As long as it pays within the credit terms, this keeps cash in the business longer.

Ratio	2026	2025	Trend
Stock turnover	6,7 times	7,4 times	Slower
Debtors collection period	63,9 days	60,8 days	Longer
Creditors payment period	58,4 days	56,4 days	Longer

Common mistakes

- Using closing stock or closing equity instead of the average in turnover and return on equity. Always average when opening and closing figures are available.
- Writing only the number in an interpretation question and forgetting the comparison, cause and consequence.
- Mixing up the debt-equity ratio (non-current liabilities) with a total-liabilities-based ratio without saying which one is used.
- Forgetting to multiply by 100 when the answer must be a percentage, or by 365 when the answer must be in days.
- Treating a very high current ratio as automatically good. Stock that cannot be sold and debtors who do not pay can still leave the business short of cash.
- Using total sales in the debtors collection period when the question says to use credit sales only.

Quick summary / cheat sheet

Group	Ratio	Ideal or benchmark
Profitability	Gross profit margin	Higher is better; compare to prior year
Profitability	Net profit margin	Higher is better
Profitability	Return on equity	Higher is better; compare to other investments
Liquidity	Current ratio	About 2 : 1
Liquidity	Acid-test ratio	About 1 : 1
Solvency	Debt-equity ratio	Lower means less risk
Efficiency	Stock turnover	Higher means faster movement
Efficiency	Debtors collection period	Shorter means faster cash inflow
Efficiency	Creditors payment period	Longer keeps cash longer, if within terms

Interpretation sentence starter: "The ratio [improved / weakened] from ... to ..., because ..., which means the business ..."

Practice questions

1. Gross profit is R450 000,00 and sales are R1 500 000,00. Calculate the gross profit margin.
2. Profit after tax is R180 000,00 and sales are R1 200 000,00. Calculate the net profit margin.
3. Current assets are R480 000,00, inventories are R160 000,00 and current liabilities are R200 000,00. Calculate the current ratio and the acid-test ratio.
4. Non-current liabilities are R400 000,00 and equity is R1 600 000,00. Calculate the debt-equity ratio.
5. Cost of sales is R900 000,00, opening stock is R120 000,00 and closing stock is R180 000,00. Calculate the stock turnover.
6. Trade receivables are R250 000,00 and credit sales are R1 825 000,00. Calculate the debtors collection period in days.
7. Trade payables are R150 000,00 and credit purchases are R1 095 000,00. Calculate the creditors payment period in days.

Answers

1. $R450\,000,00 \div R1\,500\,000,00 \times 100 = \mathbf{30,0\%}$.
2. $R180\,000,00 \div R1\,200\,000,00 \times 100 = \mathbf{15,0\%}$.
3. Current ratio = $R480\,000,00 \div R200\,000,00 = \mathbf{2,4 : 1}$. Acid-test ratio = $(R480\,000,00 - R160\,000,00) \div R200\,000,00 = R320\,000,00 \div R200\,000,00 = \mathbf{1,6 : 1}$.
4. $R400\,000,00 \div R1\,600\,000,00 = \mathbf{0,25 : 1}$.
5. Average stock = $(R120\,000,00 + R180\,000,00) \div 2 = R150\,000,00$. $R900\,000,00 \div R150\,000,00 = \mathbf{6 \text{ times}}$.
6. $R250\,000,00 \div R1\,825\,000,00 \times 365 = \mathbf{50 \text{ days}}$ (rounded).
7. $R150\,000,00 \div R1\,095\,000,00 \times 365 = \mathbf{50 \text{ days}}$ (rounded).

Official EduRank Team notes. Always cross-check with your textbook and teacher.